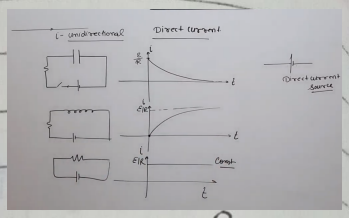
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# Alternating current

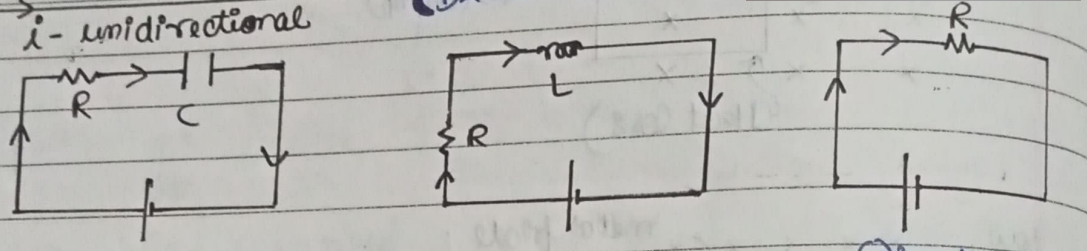
(lecture-1)



## # Direct Current :

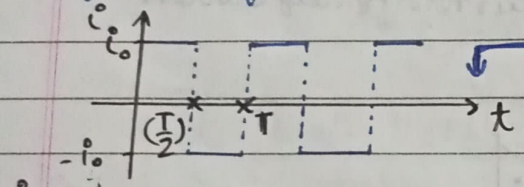
→ i - unidirectional (Direct current)

eg →

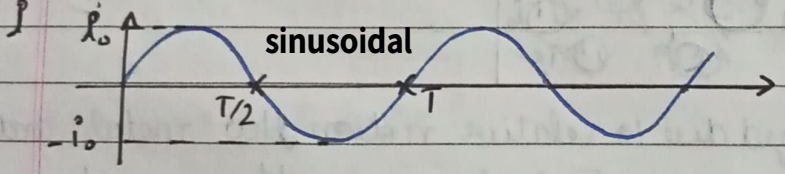



## # Alternating Current :

→ If dirⁿ of current changes with time, then it is AC.




Square wave current




$i = i_0 \sin(\omega t)$ (ex-1)

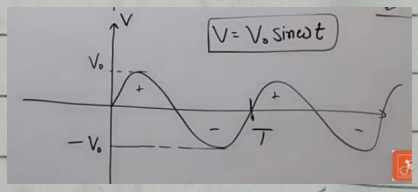
So, $i = (i_0)\sin(\omega t)$

- $i$ → current at any time
- $i_0$ → Current amplitude / Peak current
- $\omega$ → angular frequency

$T = \frac{2\pi}{\omega}$ $\quad$ $\omega = 2\pi f \Rightarrow f = \frac{\omega}{2\pi}$

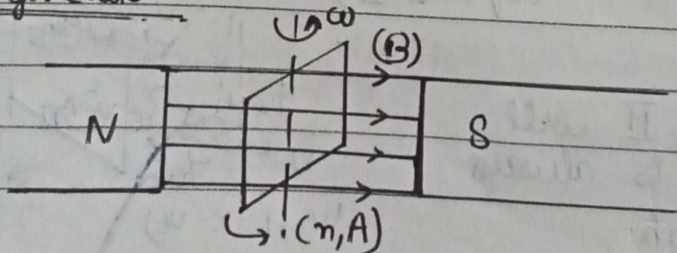
T → time taken to complete one cycle

f → no. of cycles completed per second.




(i)

## AC generator




(India : House hold electricity AC

$f = 50$ Hz

$\omega = (100)\pi$ rad/s

$\mathcal{E}_{0(\text{peak})} = 220\sqrt{2}$ volts

now, $\phi = (B)(nA)\cos\omega t$

$$\varepsilon = -\frac{d\phi}{dt}$$

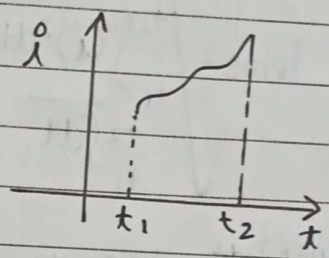
$$= (BnA\omega)\sin(\omega t)$$

$$= \varepsilon_0 \sin(\omega t) = \varepsilon_0 \sin(2\pi f t)$$

## Average current

$$i_{avg} = \frac{q}{t_2 - t_1}$$

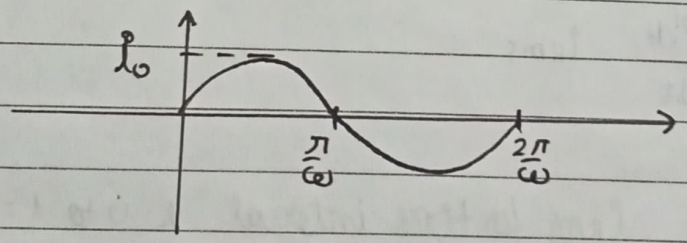
$$= \frac{\int_{t_1}^{t_2} i\,dt}{\int_{t_1}^{t_2} dt}$$




So, $i_{avg} = \dfrac{\int_{t_1}^{t_2} i\,dt}{\int_{t_1}^{t_2} dt}$

$i_{Avg} = \dfrac{\text{area under } i\text{-}t \text{ curve}}{\Delta t}$

(x-1) $i = i_0 \sin(\omega t)$




Find $i_{avg}$

(i) $t = 0$ to $t = 2\pi/\omega$

$$i = \frac{\int i\,dt}{\int dt}$$

$$= \frac{\int_0^{2\pi/\omega} i_0 \sin\omega t\,dt}{\int_0^{2\pi/\omega} dt}$$

$$= \frac{\frac{i_0}{\omega}(-\cos\omega t)_0^{2\pi/\omega}}{2\pi/\omega}$$

$2\pi/\omega = T$

$$= \frac{i_0\{-\cos 2\pi + \cos 0\}}{2\pi}$$

$$= 0$$

(ii) $t=0$ to $t=\pi/\omega$

$$= \left(\frac{2i_0}{\pi}\right)$$

## Effective value of current / RMS value

Root Mean Square

$i(t)$

$$i_{RMS} = \sqrt{\frac{\int_{t_1}^{t_2} (i)^2 dt}{\int_{t_1}^{t_2} dt}}$$

$$V(t) \quad \sqrt{\frac{\int_{t_1}^{t_2} V^2 dt}{\int_{t_1}^{t_2} dt}} = V_{RMS}$$

$i(t)$ R

$P dt = i^2 R dt$, so, $H = \int P dt \Rightarrow \Delta Q = \int_{t_1}^{t_2} i^2 R dt$ ...(i)

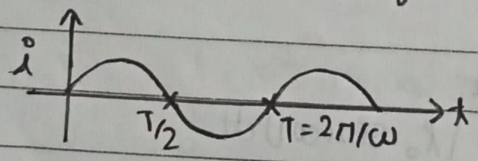
$i_c$ R

here, $\Delta Q = Pt = (i_c)^2 R (t_2 - t_1)$

equating (i) and (ii), $(i_c)^2 = \frac{\int_{t_1}^{t_2} i^2 R dt}{R(t_2 - t_1)}$

$$i_c = \sqrt{\frac{\int_{t_1}^{t_2} i^2 dt}{\int_{t_1}^{t_2} dt}} = i_{RMS}$$

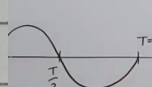
Ex-2 $i = i_0 \sin(\omega t)$ find $i_{RMS}$ for time interval $t=0$ to $t=2\pi/\omega$.




Soln

Due of current / RMS Value

$i = (i_0)\sin\omega t$




Find $i_{rms}$ for time interval $t=0$ to $t=2\pi/\omega$

$$i_{rms} = \sqrt{\frac{\int i^2 dt}{\int dt}} = \sqrt{\frac{\int_0^{2\pi/\omega} (i_0^2 \sin^2\omega t)\, dt}{\int_0^{2\pi/\omega} dt}} = \sqrt{\frac{\frac{i_0^2}{2}\int_0^{2\pi/\omega} \left(1 - \cos 2\omega t\right) dt}{2\pi/\omega}}$$

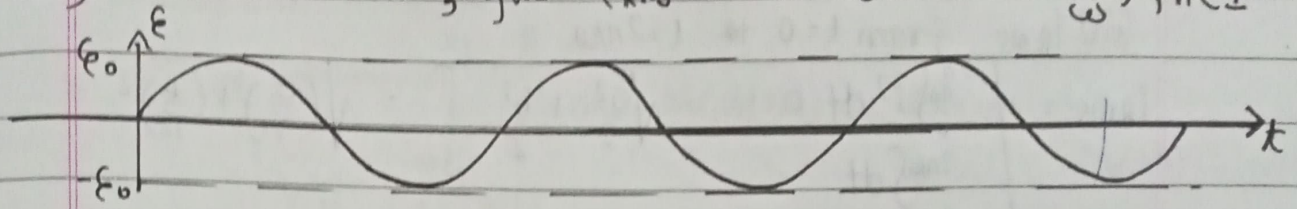
$$= \sqrt{\frac{\frac{i_0^2}{2} \cdot \frac{2\pi}{\omega}}{2\pi/\omega}} = \frac{i_0}{\sqrt{2}}$$

$$\int (1 - \cos 2\omega t)\, dt = \left(t - \frac{\sin 2\omega t}{2\omega}\right)_0^{\frac{2\pi}{\omega}}$$

$$= \left(\frac{2\pi}{\omega} - \frac{\sin 2\omega \cdot \frac{2\pi}{\omega}}{2\omega}\right) - (0 - 0)$$

$= 0.707 i_0$

Ex-3) $\mathcal{E} = \mathcal{E}_0 \sin(\omega t)$ ; find $\mathcal{E}_{RMS}$ in $t=0$ to $t = nT \left(n\frac{2\pi}{\omega}\right)$, $n \in I$

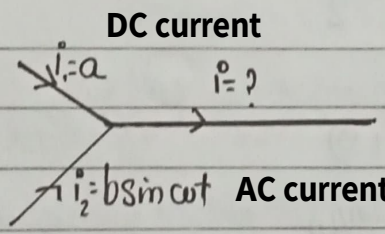



Soln $\mathcal{E}_{RMS} = \sqrt{\frac{\int_0^{nT} \mathcal{E}^2 dt}{\int_0^{nT} dt}} = \frac{\mathcal{E}_0}{\sqrt{2}}$

So, $\frac{\int_0^{nT} \mathcal{E}_0^2 \sin^2(\omega t)\, dt}{\int_0^{nT} dt} = \frac{(\mathcal{E}_0)^2}{2} \frac{\int_0^{nT} (1 - \cos 2\omega t)\, dt}{\int_0^{nT} dt}$

$= \left(\frac{\mathcal{E}_0^2}{2}\right) \frac{\left\{ \left(t - \frac{\sin 2\omega t}{2\omega}\right)_0^{nT} \right\}}{(t)_0^{nT}} = \sqrt{\frac{\mathcal{E}_0^2}{2}}$

Ex-4) find $i_{RMS}$ $t=0$ to $t = \frac{2\pi}{\omega}$




$\int i^2 dt$

$= \int (a + b\sin\omega t)^2 dt$

$= a^2 \int_0^{\frac{2\pi}{\omega}} dt + \frac{b^2}{2} \int_0^{\frac{2\pi}{\omega}} (1 - \cos 2\omega t)\, dt + 2ab \int_0^{\frac{2\pi}{\omega}} \sin(\omega t)\, dt$

$= a^2 (t)_0^{2\pi/\omega} + \frac{b^2}{2} \left(t - \frac{\sin 2\omega t}{2}\right)_0^{\frac{2\pi}{\omega}} + 2ab \left(-\frac{\cos\omega t}{\omega}\right)_0^{2\pi/\omega}$

$= a^2 (2\pi/\omega) + b^2/2\, (2\pi/\omega)$

$\sqrt{\frac{\int_0^{2\pi/\omega} i^2 dt}{\int_0^{\frac{2\pi}{\omega}} dt}} = i_{RMS} = \sqrt{a^2 + \frac{b^2}{2}}$

$= \sqrt{(i_1)^2 + (i_{RMS})^2}$ (of 2)

Ex-5) $i = a\sin(\omega t) + b\cos(\omega t)$

find $i_{RMS}$ from $t=0$ to $t = 2\pi/\omega$

**# Method-1:**

$$i_{RMS} = \sqrt{\frac{\int_0^{2\pi/\omega} i^2\,dt}{\int_0^{2\pi/\omega} dt}} = \sqrt{\frac{a^2}{2} + \frac{b^2}{2}} = \sqrt{\left(\frac{a}{\sqrt{2}}\right)^2 + \left(\frac{b}{\sqrt{2}}\right)^2}$$

$$\int_0^{2\pi/\omega} i^2 dt = \int \left(a^2\sin^2\omega t + b^2\cos^2\omega t + 2ab\sin(\omega t)\cos(\omega t)\right) dt$$

$$= \int_0^{2\pi/\omega} \left( \frac{a^2}{2}(1-\cos 2\omega t) + \frac{b^2}{2}(1+\cos 2\omega t) + 2ab\frac{\sin 2\omega t}{2} \right) dt$$

$$= \int_0^{2\pi/\omega} \left(\frac{a^2}{2} + \frac{b^2}{2}\right) dt$$

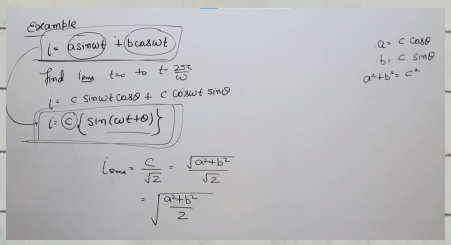
**in 0 to 2pi, integration of all trigonometric terms will give zero**

**# Method-2:**

Ans, let $a = C\cos(\theta)$, $b = C\sin(\theta)$

$$i = C(\sin(\omega t + \theta))$$

$$i_{rms} = \frac{C}{\sqrt{2}} = \sqrt{\frac{a^2+b^2}{2}}$$




Ex-6) $i = a\sin(\omega t) + b\sin(\omega t + \theta)$

find $i_{RMS}$ $t = 0$ to $t = 2\pi/\omega$

Ex $i = a\sin\omega t + b\sin(\omega t+\theta)$

find $i_{rms}$ $t=0$ से $t = \frac{2\pi}{\omega}$

$2\sin C\sin D = \cos(C-D) - \cos(C+D)$

$\cos(C+D) = \cos C\cos D - \sin C\sin D$

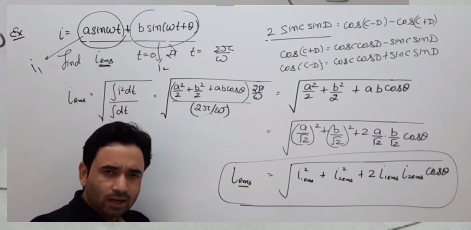
$\cos(C-D) = \cos C\cos D + \sin C\sin D$

$$i_{Rms} = \sqrt{\frac{\int i^2 dt}{\int dt}} = \sqrt{\frac{\left(\frac{a^2}{2}+\frac{b^2}{2}+ab\cos\theta\right)\frac{2\pi}{\omega}}{(2\pi/\omega)}} = \sqrt{\frac{a^2}{2}+\frac{b^2}{2}+ab\cos\theta}$$

$$\int i^2 dt = \int_{2\pi/\omega} \left[a^2\sin^2\omega t + b^2\sin^2(\omega t+\theta) + 2ab\sin(\omega t+\theta)\sin\omega t\right] dt$$

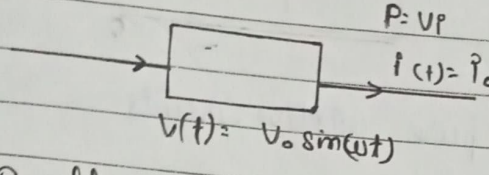
$$= \int_0^{2\pi/\omega} \left[\frac{a^2}{2}(1-\cos 2\omega t) + \frac{b^2}{2}\{1-\cos(2\omega t+2\theta)\} + ab\{\cos\theta - \cos(2\omega t+\theta)\}\right] dt$$

$$= \left(\frac{a^2}{2}t + \frac{b^2}{2}t + ab\cos\theta\, t\right)_0^{2\pi/\omega}$$



(lecture -2)

## Power of source or load

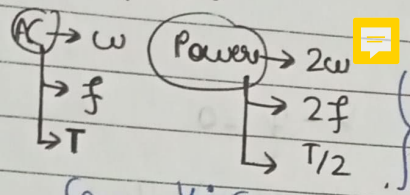



⇒ Power = VI

$= V_0 i_0 \sin(\omega t+\theta)\sin(\omega t)$

$= \frac{V_0 i_0}{2}(\cos\theta - \cos(\theta+2\omega t))$

θ = phase diff between i and V
or (power factor) angle




$P_{avg} = \frac{\int P\,dt}{\int dt} = \frac{\frac{V_0 i_0}{2}\int(\cos\theta\,dt - \cos(2\omega t+\theta)\,dt)}{2\pi/\omega}$

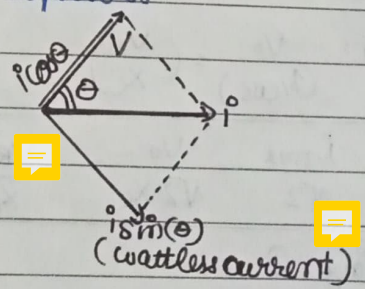
$= \frac{V_0 i_0}{2(2\pi/\omega)}\cos\theta(2\pi/\omega) = \left(\frac{V_0 i_0}{2}\right)(\cos\theta)$ ← power factor

$= \left(\frac{V_0}{\sqrt{2}}\right)\cdot\left(\frac{i_0}{\sqrt{2}}\right)(\cos\theta) = (V_{rms})\cdot(i_{rms})\cdot(\text{power factor})$

**Phasors** :⇒ Imaginary vectors representing sinusoidally varying voltage or current are known as phasors.

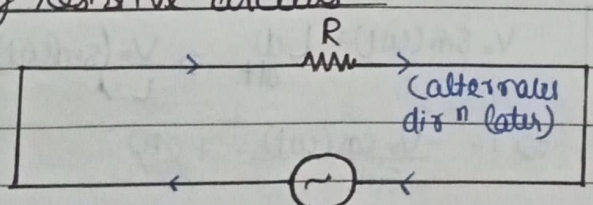
$i = i_0 \sin(\omega t)$

$V = V_0 \sin(\omega t+\theta)$




## AC circuits

### (I) Purely resistive circuits




$P_{avg} = (V_{rms})(i_{rms})\cos 0^\circ = \left(\frac{\varepsilon_0^2}{2R}\right)$

$\varepsilon - iR = 0$

$\varepsilon = iR \Rightarrow i = \frac{\varepsilon}{R}\sin\omega t$

$i = i_0 \sin\omega t$

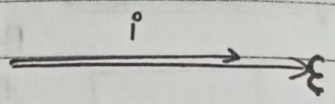
phase diff b/w voltage and current = 0

power factor = cos(0) = 1

$i_{max} = \frac{\varepsilon}{R}$, $i_{RMS} = \frac{\varepsilon_{RMS}}{R}$

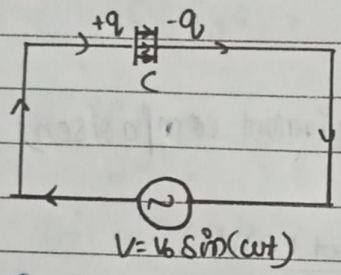
Heat = $(P_{avg}) \times t$

Phasor: length – Amplitude
dir$^n$ – Phase




'V' and 'i' are in phase in pure resistive circuits

## (II) Purely capacitive circuit




$V - \frac{q}{C} = 0$

So, $V_0 \sin(\omega t) = \frac{q}{C}$

So, $q = C V_0 \sin(\omega t)$

$i = \frac{dq}{dt} = C V_0 (\omega) \sin(\omega t)$

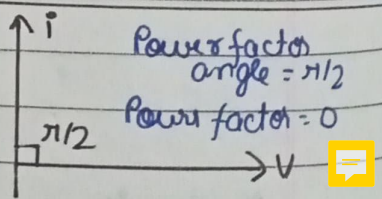
$q = C V_0 \sin(\omega t) \qquad i = C V_0 \omega \cos(\omega t)$

$i = \left(\frac{V_0}{1/\omega C}\right) \sin(\omega t + \pi/2) = \frac{V_0}{(1/\omega C)} \sin(\omega t + \pi/2)$

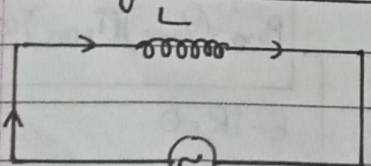
$i_{max} = \frac{V_0}{(1/\omega C)} = \left(\frac{V_0}{X_C}\right)$ $\qquad X_C$ = capacitive reactance $= 1/\omega C = 1/2\pi fC$

$i_{RMS} = \frac{i_{max}}{\sqrt{2}} = \frac{V_0}{\sqrt{2} X_C} = \frac{V_{RMS}}{X_C}$

$P_{avg} = 0$




## (III) Purely Inductive circuits




by applying KVL,

$V = V_0 \sin(\omega t) = L \frac{di}{dt} \Rightarrow \frac{V_0}{L} \int \sin(\omega t) dt = \int di$

$\Rightarrow i = \frac{-V_0 \cos(\omega t)}{\omega L} + C$

So, $i = \frac{V_0 \sin(\omega t - \pi/2)}{\omega L} = \frac{V_0 \sin(\omega t - \pi/2)}{\omega L}$ $\qquad (\omega L) = X_L$

$\Rightarrow$ Inductive reactance

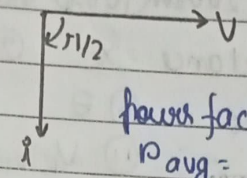
$i = \frac{V_0}{X_L} \sin(\omega t - \pi/2)$

$X_L \uparrow$ ; $i \downarrow$

$i_{max} = i_0 = \frac{V_0}{X_L} = \left(\frac{V_0}{\omega L}\right)$

$i_{RMS} = \frac{V_0}{\sqrt{2} X_L} = \frac{V_0}{\sqrt{2}(\omega L)}$

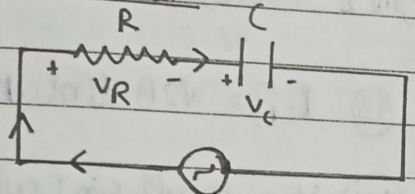
current in inductor is lagging the voltage across inductor by phase $\frac{\pi}{2}$.




power factor = 0

$P_{avg} = 0$

(IV) R-C circuit

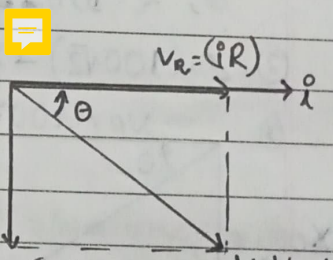



$V = V_0 \sin(\omega t)$

KVL, $V - V_R - V_C = 0$

$V = V_R + V_C$

$V_0 \sin(\omega t) = (iR) + \frac{q}{C}$ $\Big\}$ $i = \frac{dq}{dt}$




$V = i\sqrt{R^2 + X_C^2}$

$Z$ = impedance $= \sqrt{R^2 + X_C^2}$ (ohm)

hence, $V = IZ$

$\frac{q}{it}$ $\tan\theta = \frac{V_C}{V_R} = \frac{iX_C}{iR} = \frac{X_C}{R}$ $= \sqrt{R}$

Power factor $\cos\theta = \frac{V_R}{V} = \frac{iR}{iZ} = R/Z$

now, $i_{max} = \frac{V_{max}}{Z} = \frac{V_0}{\sqrt{R^2 + X^2}}$

$i_{RMS} = \frac{V_0}{\sqrt{2(R^2 + X^2)}} = \frac{V_{RMS}}{Z}$ $\Big\}$ $P_{avg} = i(RMS) \cdot V_{(RMS)} \cdot \cos(\theta)$

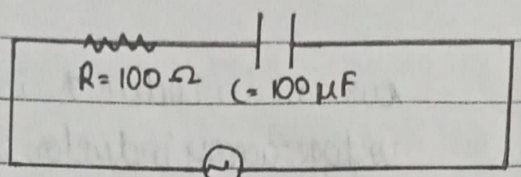
$= \frac{(V_{RMS})^2}{Z} \times \frac{R}{Z}$

$= \left(\frac{V_{RMS}}{Z}\right)^2 \cdot R = (i_{RMS}^2 R)$

$i(t) = \frac{V_0}{Z} \sin\left(\omega t + \tan^{-1}\left(\frac{X_C}{R}\right)\right)$

$= \frac{V_0}{Z} \sin\left(\omega t + \cos^{-1}\left(\frac{R}{Z}\right)\right)$

Ex-7)



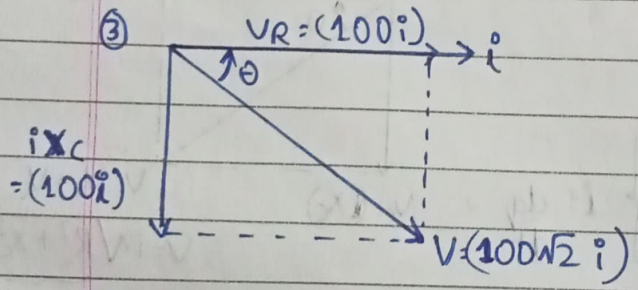

find ① capacitive reactance = $X_C$ ② Impedance = Z ③ Draw phasor
④ $i_{max}$ ⑤ $i_{RMS}$ ⑥ θ ⑦ Power factor ⑧ Avg Power
⑨ $V_{R\,max}$ ⑩ $V_{C\,max}$ ⑪ $V_{R\,RMS}$ ⑫ $V_{C\,RMS}$ ⑬ $i_{(t)}$
⑭ $V_{R(t)}$ ⑮ $V_{C(t)}$

Soln ① $X_C = 1/(\omega C) = 1/(100)(100) = (10^{-4})\,M\Omega = (10)^2\,\Omega = 100\,\Omega$

So, $X_C = 100\,\Omega$

② $Z = (100\sqrt{2})\,\Omega$

③




④ $i_{max} = \dfrac{V_{max}}{Z} = \dfrac{(200V)}{(100\sqrt{2})\,\Omega} = \sqrt{2}\,(A)$

⑤ $i_{RMS} = \left(\dfrac{\sqrt{2}}{\sqrt{2}}\right) A =$ 1 Ampere

⑥ $\theta = 45°$ ⑦ Power factor $= \cos(\theta) = \dfrac{1}{\sqrt{2}} = \left(\dfrac{R}{Z}\right)$

⑧ $P_{avg} = \left(\dfrac{200}{\sqrt{2}}\right) \times (1) \times \cos(45°) = 100$ watt

⑨ $V_{R\,max} = (i_{max})\,R = (\sqrt{2})\,(100) = (100\sqrt{2})$ volts

⑩ $V_{C\,max} = (i_{max})(X_C) = 100\sqrt{2}$ volts

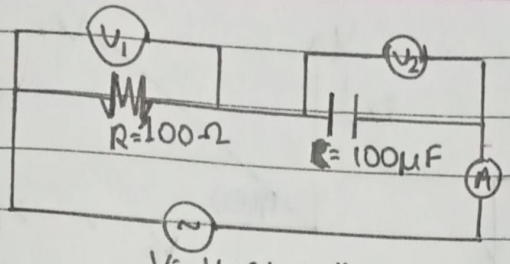
⑪ $V_{R\,RMS} = 100V$ ⑫ $V_{C\,RMS} = 100V$

⑬ $i_{(t)} = \sqrt{2}A \sin(\omega t + \frac{\pi}{4})$

⑭ $V_{R_{(t)}} = 100\sqrt{2} \sin(\omega t + \frac{\pi}{4})$

⑮ $V_{C_{(t)}} = 100\sqrt{2} \sin(\omega t - \frac{\pi}{4})$

Ex-8)




Ideal AC ammeter A, and voltmeter $V_1, V_2$. find their readings.

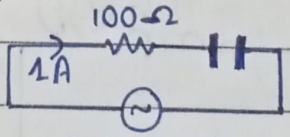
Sol$^n$ (AC) measuring instruments measure RMS values.

$(V_1) = 100$ Volts
$(V_2) = 100$ V
$(A) = 1$ Ampere
} effective readings

(Lecture-3)

Ex-9) A bulb is rated as 100 W, 100 Volts. What capacitance should be applied in series to the bulb so that it provides rated power with AC source 200V, 50Hz.
(until and unless stated, we consider them RMS value)

Sol$^n$ $(R) = \frac{V^2}{P} = 100\Omega$




$V = 200\sqrt{2}\sin(\frac{50}{100\pi}t)$

$i_{RMS} = 1A = \left(\frac{V_{RMS}}{Z}\right)$

So, $Z = \frac{200V}{1A}$

$\sqrt{R^2 + (X_C)^2} = (200)$

$R^2 + (X_C)^2 = 40000$

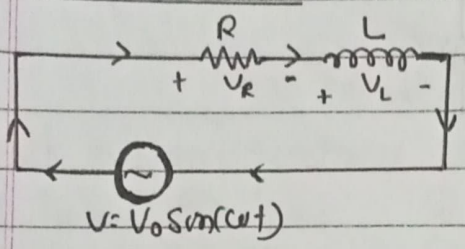
$(X_C)^2 = 30000 \Rightarrow X_C = 100\sqrt{3}\ (\Omega)$

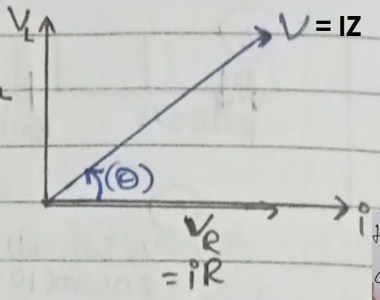
So, $X_C = \frac{1}{(\omega)(C)}$

$C = \frac{1}{(100\pi)(100\sqrt{3})}$

$C = \frac{100}{\pi\sqrt{3}}\ \mu F$

⑤ L-R circuit




$\tan\theta = \frac{iX_L}{iR} = \frac{X_L}{R}$

$\cos\theta = \frac{V_R}{V} = \frac{IR}{IZ} = R/Z$

$\theta = \tan^{-1}\left(\frac{X_L}{R}\right) = \cos^{-1}\left(\frac{R}{Z}\right)$

Power factor $= (R/Z)$

$i_{max} = i_0 = \frac{V_0}{Z}$

$i_{RMS} = \left(\frac{V_{RMS}}{Z}\right)$

} In inductive circuit, current will lag V

$Z$ = Impedance $= \sqrt{R^2 + X_L^2}$

$= \sqrt{(\omega L)^2 + R^2}$

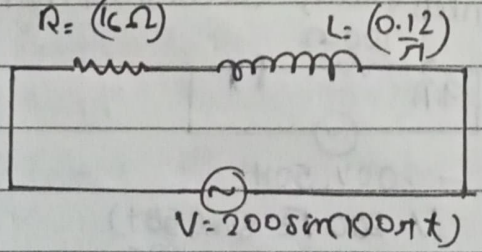
$= \sqrt{(2\pi f L)^2 + R^2}$

$i_{(t)} = \frac{V_0}{Z} \sin\left(\omega t - \tan^{-1}\left(\frac{X_L}{R}\right)\right)$

$V_{R(t)} = (i_{(t)} R) = \left(\frac{V_0 R}{Z}\right) \sin\left(\omega t - \cos^{-1}\left(\frac{R}{Z}\right)\right)$

$V_{L(t)} = \left(\frac{V_0}{Z} X_L\right) \sin\left(\omega t + \frac{\pi}{2} - \theta\right)$

Ex.10)




find

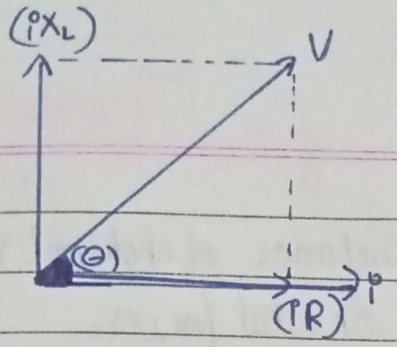
① Inductive reactance ② Impedance ③ draw phasor ④ $\theta$

⑤ Power factor ⑥ $i_{max}$ ⑦ Pavg ⑧ $(V_R)_{max}$ ⑨ $(V_L)_{max}$

⑩ $i(t)$ ⑪ $V_R(t)$ ⑫ $V_L(t)$

Soln ① $X_L = (\omega L) = \left(100\pi \times \frac{0.12}{\pi}\right) = 12\,\Omega$

② $Z = \sqrt{R^2 + (X_L)^2} = \sqrt{256 + 144} = 20\,\Omega$




③

④ $\theta = \tan^{-1}(12/16) = (37°)$

⑤ power factor = $\cos(\theta) = \cancel{37}\ (4/5)$

⑥ $i_{max} = \frac{V_0}{2} = (10A)$

⑦ $i_{rms} = (5\sqrt{2})A$ , $P_{avg.} = (5\sqrt{2})\left(\frac{200}{\sqrt{2}}\right) \times \frac{4}{5} = 800$ Watt

⑧ $V_{R\ max} = (160V)$

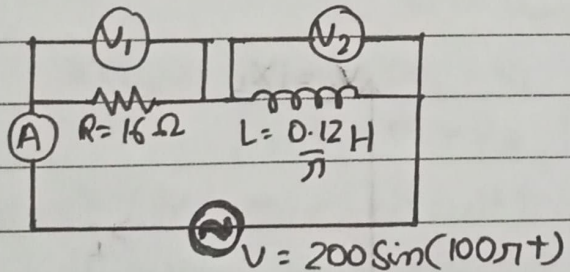
⑨ $(V_L)_{max} = (120V)$

⑩ $i(t) = 10 \sin(100\pi t - 37°)$

⑪ $(V_R)(t) = 160 \sin(100\pi t - 37°)$

⑫ $(V_L)(t) = 120 \sin(100\pi t + 53°)$

(ex-11)



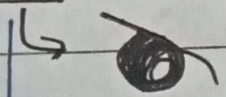

find reading of ideal ammeter and volt meters.

Soln $A = 5\sqrt{2}$ Ampere

$V_1 = 80\sqrt{2}$ volts

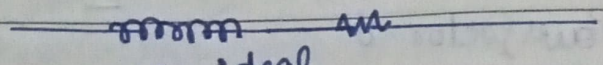
$V_2 = 60\sqrt{2}$ volts

## Choke coil

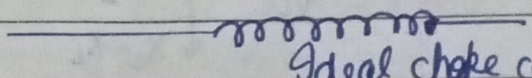


↳ A wire having many loops, thus Inductance (say L)

hence, it is an inductor

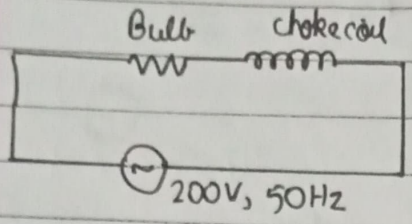


non-ideal



Ideal choke coil

Ex-12) Bulb: rated 100V, 200Watt — find self inductance of choke coil so that bulb works at rated power.




Sol$^n$ $R_{bulb} = \left(\frac{V^2}{P}\right) = \frac{(10^4)}{200} = 50\,\Omega$

$(R = 50\,\Omega)$

power $= (i_{rms})^2 R \Rightarrow i_{rms} = 2A$

$i_{RMS} = \frac{V_{RMS}}{Z}$

$R^2 + (X_L)^2 = (100)^2$

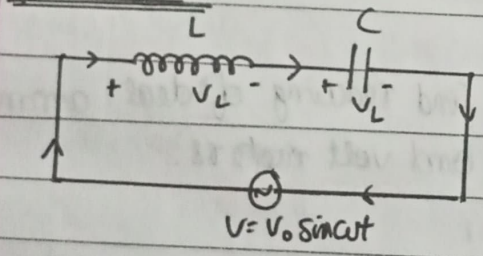
$(X_L)^2 = (150)(50)$

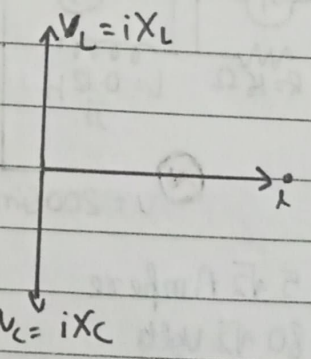
$X_L = (50\sqrt{3})$

$\omega L = 50\sqrt{3} \Rightarrow (2\pi)(50)\,L = 50\sqrt{3}$

$L = \frac{\sqrt{3}}{2\pi}$ Henry

(VI) L-C circuit




KVL, $V - V_L - V_C = 0$

Case 1, $X_L > X_C$

$\omega > \frac{1}{\sqrt{LC}}$

$i(X_L - X_C) = Zi$

$\pi/2$

$\rightarrow i$

i is lagging V by $\theta = \pi/2$ ↳ Inductive circuit

} Power factor angle $= \pi/2$, Power factor $= 0$, $P_{avg} = 0$

Case-2, $X_L < X_C$

$\omega < \frac{1}{\sqrt{LC}}$

$\pi/2$

$i(X_C - X_L)$

$\rightarrow i$

} $i_{RMS} = \frac{V_{RMS}}{Z}$

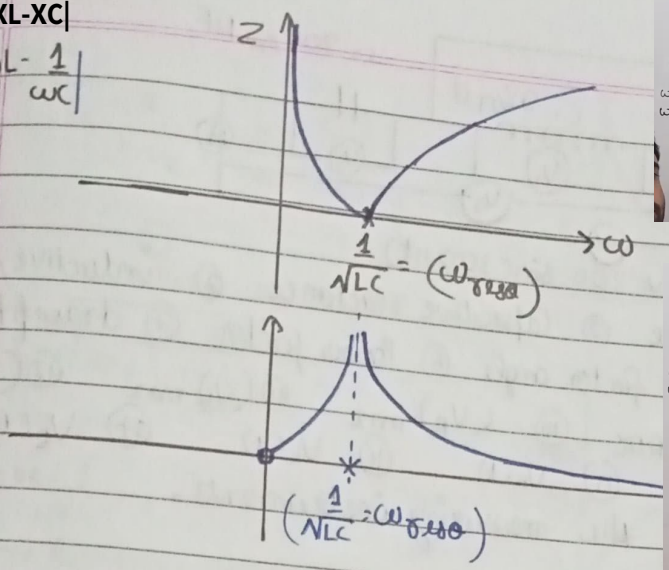
i is leading V by $\theta = \pi/2$ ↳ Capacitive circuit

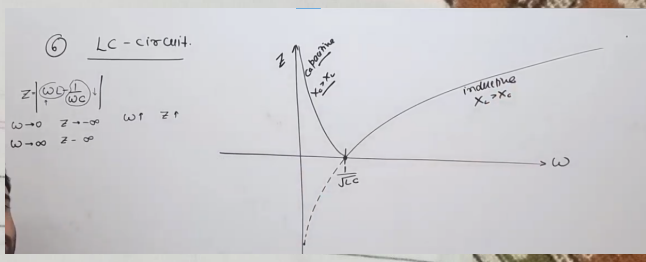
} Power factor $= 0$, $P_{avg} = 0$

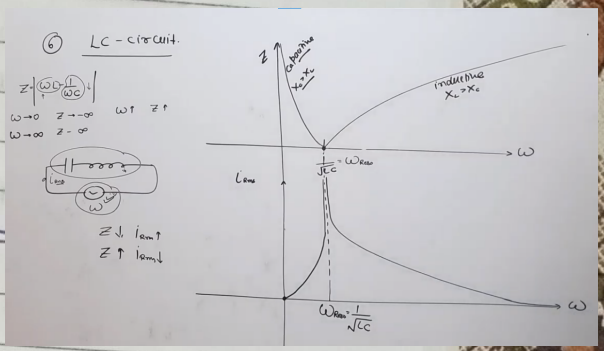
thus,

$Z = |X_L - X_C|$

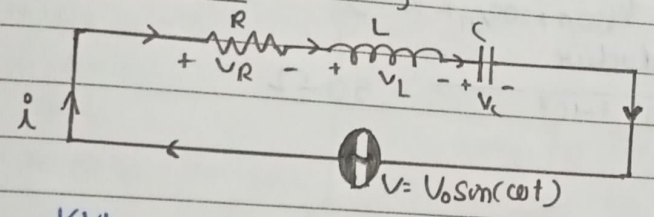
$Z = \left|\omega L - \frac{1}{\omega C}\right|$

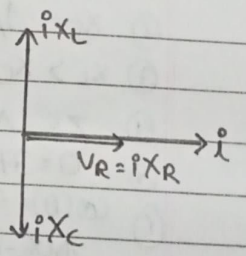



## ⑦ RLC circuit (lecture-4)




KVL, $V - V_R - V_L - V_C = 0$

$$\boxed{V = V_L + V_C + V_R}$$

**Case-1) (Inductive circuit)**

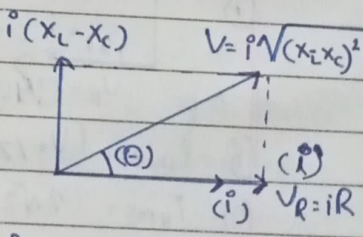
$V_L > V_C \Rightarrow X_L > X_C \Rightarrow \omega > \frac{1}{\sqrt{LC}}$

Z = impedance

$= \sqrt{(X_L - X_C)^2 + R^2}$

Power factor angle = θ

Power factor = $\cos(\theta) = (V_R/V) = (R/Z)$




$i_{max} = V_0/Z$

$P_{avg} = (i_{rms})(V_{rms})\cos\theta$

**Case-2) (Capacitive circuit)**

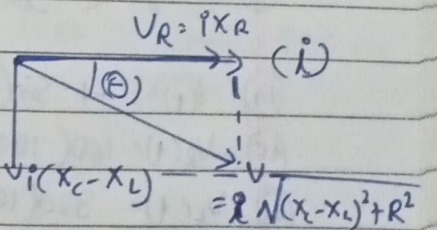
$V_C > V_L \Rightarrow \left(\omega < \frac{1}{\sqrt{LC}}\right)$

Z = Impedance

$= \sqrt{(X_C - X_L)^2 + R^2}$

Power factor angle = θ

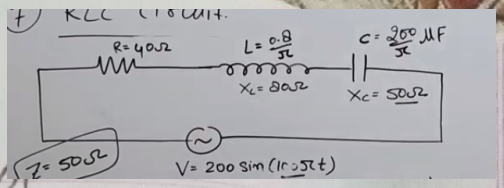
Power factor = $\cos(\theta) = R/Z = (R/Z)$



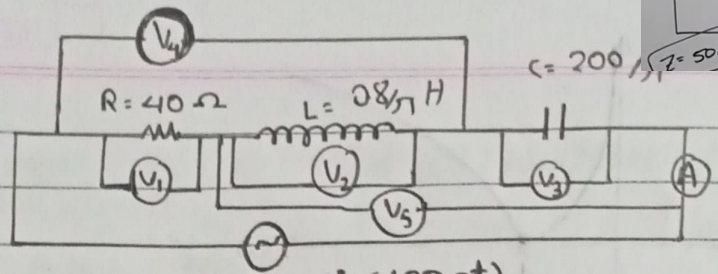

$i_{max} = V_0/Z$

$P_{avg} = (i_{rms})(V_{rms})\cos(\theta)$

$= (i_{rms})^2 R = \frac{(V_{rms})^2}{R}$




Ex-13)




Find $V = 200 \sin(100\pi t)$

① Inductive reactance ② Capacitive reactance ③ inductive/capacitive
④ Impedance ⑤ power factor angle ⑥ Power factor ⑦ draw phasor
⑧ $i_{max}$ ⑨ $i_{RMS}$ ⑩ $(V_R)_{max}$ ⑪ $(V_L)_{max}$ ⑫ $(V_C)_{max}$
⑬ $P_{avg}$ ⑭ $i(t)$ ⑮ $V_R(t)$ ⑯ $V_L(t)$ ⑰ $V_C(t)$
⑱ find readings of the measuring instruments.

Sol$^n$

① $X_L = (\omega)L = (100\pi \times 0.8/\pi) = (80)\Omega$

② $X_C = 1/\omega C = 1/(100\pi \times 200/\pi \times 10^{-6}) = (50)\Omega$

③ $X_L > X_C$, Inductive

④ $Z = \sqrt{(80-50)^2 + 40^2} = 50\Omega$

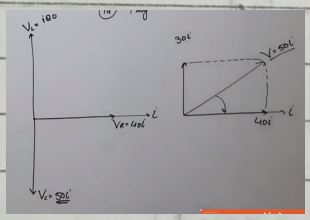
⑤ $\theta = 37°$

⑥ $\cos(\theta) = 4/5$

⑦ $(X_L - X_C)i = 30i$

$37°$

$V_R = (i40)$




Ex-14)

⑧ $i_{max} = (V_0/Z) = (200)/50 = 4A$

⑨ $i_{RMS} = 2\sqrt{2}$ Ampere

⑩ $(V_R)_{max} = (i_{max} \cdot R) = 160$ Volts

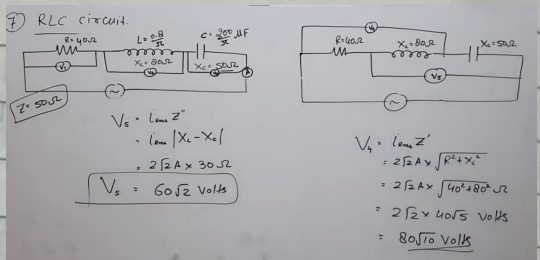
⑪ $(V_L)_{max} = (i_{max} \cdot X_L) = 320$ volts

⑫ $(V_C)_{max} = (i_{max} \cdot X_C) = 200$ volts

Sol$^n$

⑬ $P_{avg} = (2\sqrt{2})\left(\frac{200}{\sqrt{2}}\right)\left(\frac{4}{5}\right) = 320$ Watt

⑦ RLC circuit




⑭ $i(t) = 4 \sin(100\pi t - 37°)$

⑮ $V_R(t) = 160(100\pi t - 37°)$

⑯ $V_L(t) = 320(100\pi t + 53°)$

⑰ $V_C(t) = 200(100\pi t - 127°)$

⑱ A ⇒ $2\sqrt{2}$ Ampere; $V_1 \Rightarrow 80\sqrt{2}$ Volts; $V_2 \Rightarrow 160\sqrt{2}$ volts

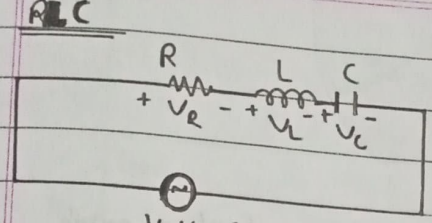
$V_3 = 100\sqrt{2}$ volts; $V_4 = (i_{rms})Z' = (2\sqrt{2}A\sqrt{40^2+80^2}) = 80\sqrt{10}$ volts

$V_5 = (i_{RMS})(|X_L - X_C|) = 60\sqrt{2}$ volts

inductive / capacitive
draw phasors
(12) $(i)_{rms}$
$V_C(t)$

DATE / / PAGE
NOTEBOOK

## RLC




$V = V_0 \sin(\omega t)$

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

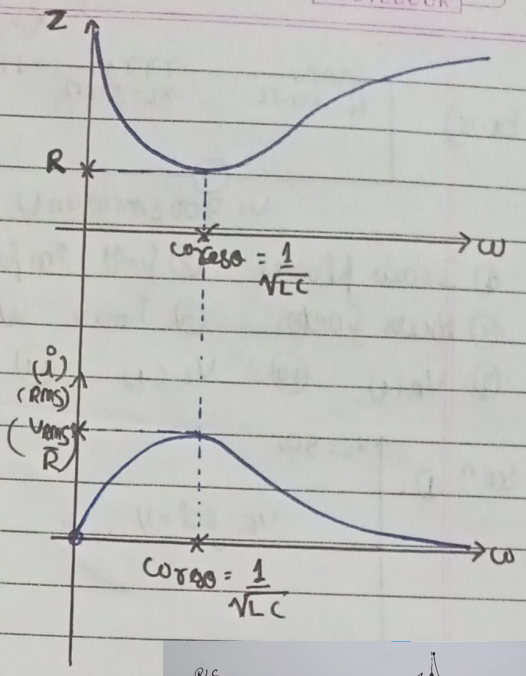
$= \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$

$\omega \to 0, Z \to \infty$

$\omega_{reso} = \frac{1}{\sqrt{LC}}$, $Z_{min} = R$, $\omega L = \frac{1}{\omega C}$

↳ Resonance angular frequency

$\omega \to \infty, Z \to \infty$




**# Resonance :**

$i X_L = V_L$

$V = V_R + V_L + V_C$

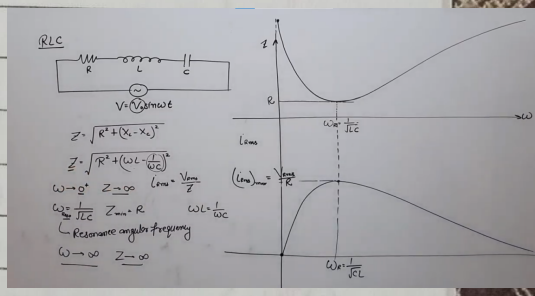
$\omega = \frac{1}{\sqrt{LC}}$

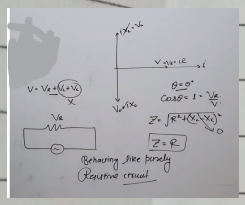
$V = V_R = iR$

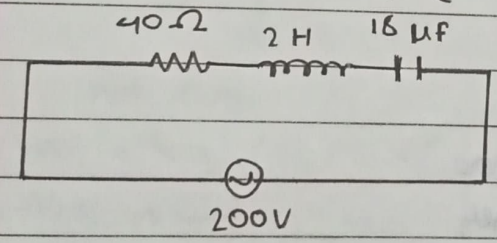
$X_L = X_C \Rightarrow V_L = V_C$

$i X_C = V_C$

$(Z = R)$



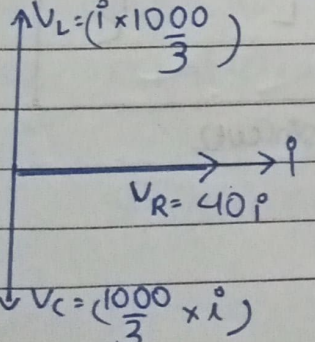

(Ex-14) find angular frequency of AC source so that system is at resonance. Also draw phasors

Soln $\omega L = \frac{1}{\omega C} \Rightarrow \omega^2 = \frac{1}{(2)(18) \times 10^{-6}}$

So, $\omega_{(reso)} = (\frac{500}{3})$ rad/s

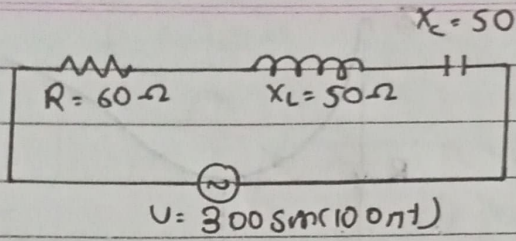
now, $\omega = 2\pi f$

$f_{reso} = \frac{\omega_{reso}}{2\pi} = \frac{500}{6\pi}$ Hz



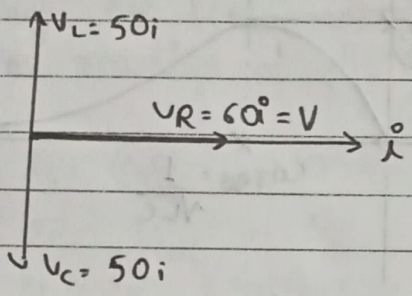

80 V to volts

Ex-15)




① Draw phasor ② find impedance ③ Power factor angle
④ Power factor ⑤ $i_{max}$ ⑥ $(V_L)_{max}$ ⑦ $(V_C)_{max}$ ⑧ $i(t)$
⑨ $V_R(t)$ ⑩ $V_L(t)$ ⑪ $V_C(t)$

Soln ①




② $Z = 60\Omega$

③ $\theta = 0^\circ$

④ Power factor = $\cos(0^\circ) = 1$

⑤ $i_{max} = (300/60) = 5$ Ampere

⑥ $(V_L)_{max} = 250$ Volts

⑦ $(V_C)_{max} = 250$ Volts

⑧ $i(t) = 5\sin(100\pi t)$ Ampere

⑨ $V_R(t) = 300\sin(100\pi t)$ Volts

⑩ $V_L(t) = 250\sin(100\pi t + \pi/2)$ Volts

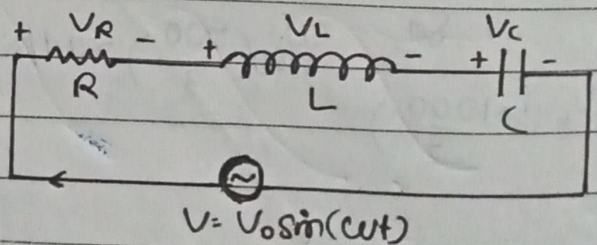
⑪ $V_C(t) = 250\sin(100\pi t - \pi/2)$ Volts

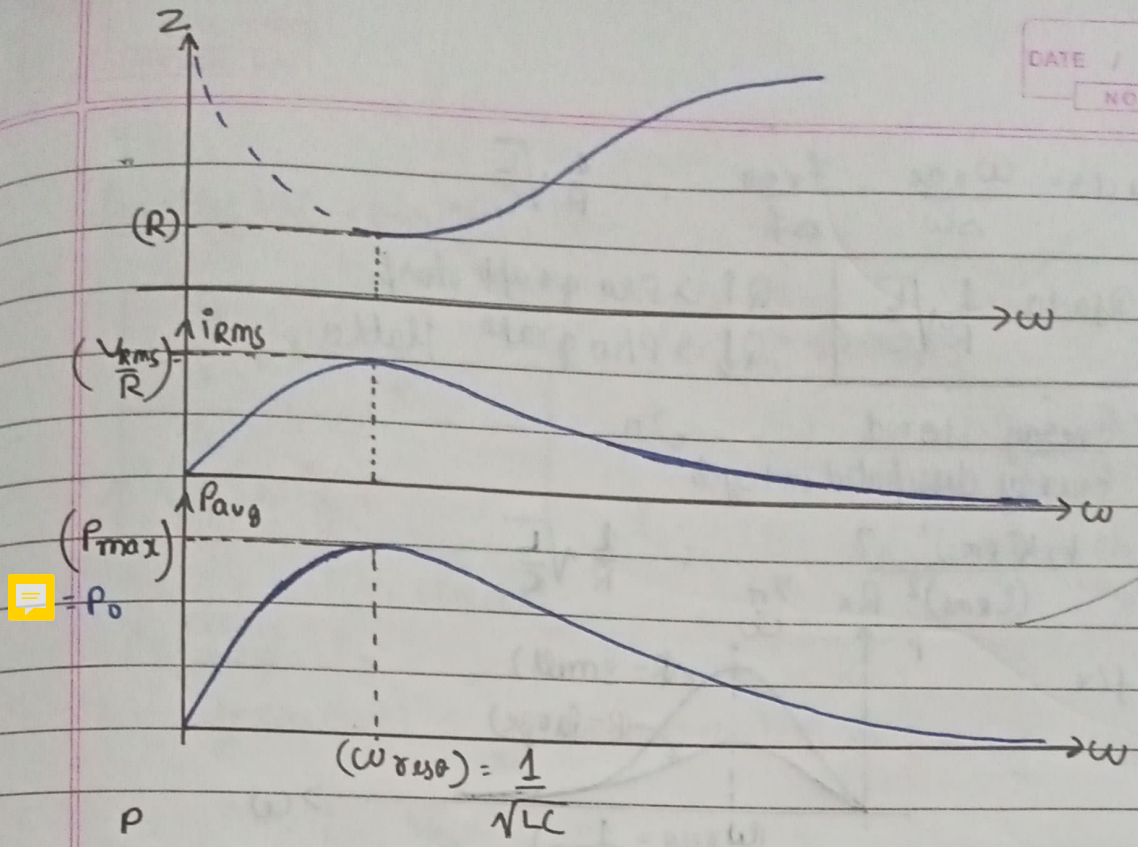
## Quality factor (lecture - 5)

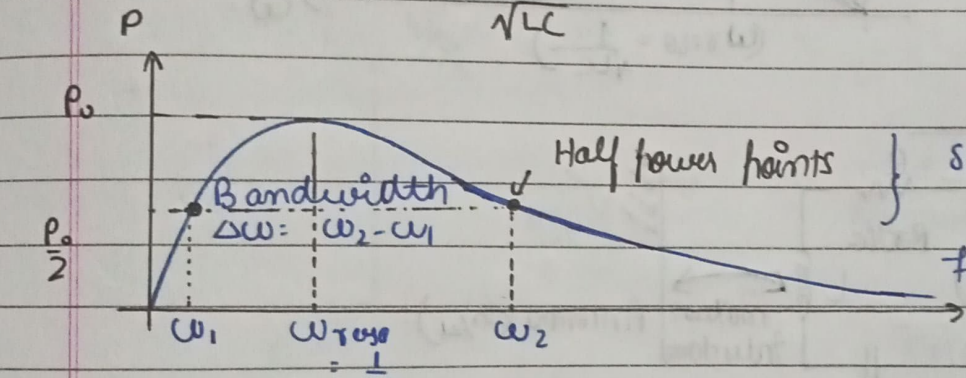
$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$i_{RMS} = \left(\frac{V_{RMS}}{Z}\right)$$

$$P_{avg} = (i^2_{RMS})(R)$$





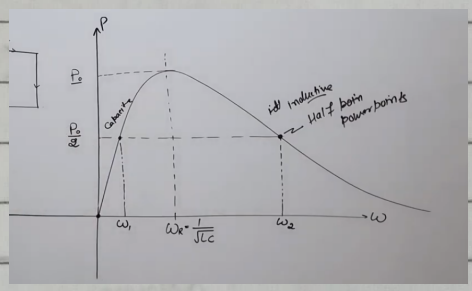

Half power points } sharp graph, $\Delta\omega$ small, $R \to$ small

flat : $\Delta\omega$ - large, $R \to$ large

$P_0 \to P_0/2$

$i_{RMS} = i_0 \to (i_0/\sqrt{2})$

$i_0 = \frac{V_{RMS}}{R} \to \frac{i_0}{\sqrt{2}} = \frac{V_{RMS}}{Z}$ } $Z = (\sqrt{2})R$




$P_0/2$, $Z = \sqrt{2}R$

$\sqrt{R^2 + (X_L - X_C)^2} = \sqrt{2}R$

Case (i)

$X_C - X_L = R$

$1/\omega C - \omega L = R$

$\omega_1 = \frac{-RC + \sqrt{R^2C^2 + 4LC}}{2LC}$

Case (ii)

$X_L - X_C = R$

$\omega L - 1/\omega C = R$

$\omega_2 = \frac{RC + \sqrt{R^2C^2 + 4LC}}{2LC}$

So, $\left(\Delta\omega = \frac{R}{L}\right)$

Case (i) (ii)

$X_C - X_L = R$

$\frac{1}{\omega C} - \omega L = R$

$1 - \omega^2 LC = \omega RC$

$\omega^2 LC + \omega RC - 1 = 0$

$\omega = \frac{-RC \pm \sqrt{R^2C^2 + 4LC}}{2LC}$

Q-factor = $\frac{\omega_{reso}}{\Delta\omega} = \frac{f_{reso}}{\Delta f} = \frac{1}{R}\sqrt{\frac{L}{C}}$

**Qfactor = $\frac{1}{R}\sqrt{\frac{L}{C}}$**

$Q\uparrow \Rightarrow$ P/ω graph sharp

$Q\downarrow \Rightarrow$ P/ω graph flatter

$Q = \frac{\text{Energy stored}}{\text{Energy dissipated in 1 cycle}} \times 2\pi$

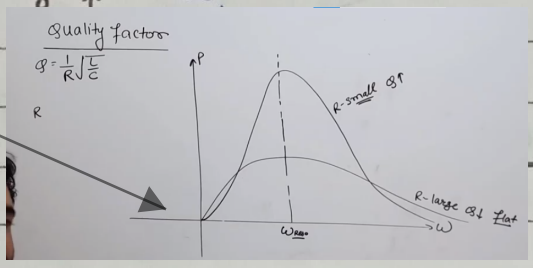
$= \frac{\frac{1}{2}L(i_{rms})^2 \times 2}{(i_{rms})^2 R \times \frac{2\pi}{\omega}} = \frac{1}{R}\sqrt{\frac{L}{C}}$

$Q = \frac{\text{Energy stored}}{\text{Energy dissipated in 1 cycle}} \times 2\pi$

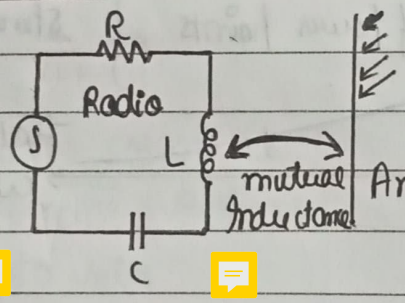
$Q = \frac{\frac{1}{2}L i_{rms}^2 \times 2}{i_{rms}^2 R \times \frac{2\pi}{\omega}} \times 2\pi$

$\therefore \frac{L}{R}\omega = \frac{L}{R} \times \frac{1}{\sqrt{LC}} = \frac{1}{R}\sqrt{\frac{L}{C}}$

Keeping L, C fix

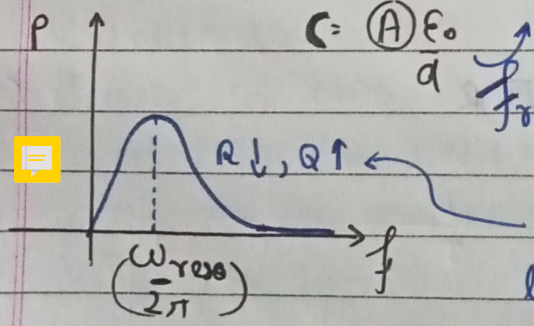



## Tuning of radio

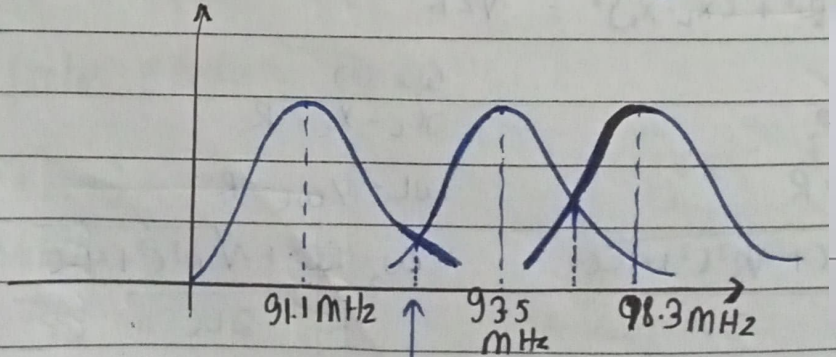



$C = \frac{A\varepsilon_0}{d}$ → $\omega_{reso} = \frac{1}{\sqrt{LC}}$

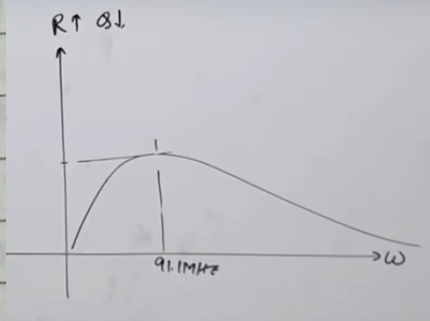
$f_{reso} = \left(\frac{\omega_{reso}}{2\pi}\right) \propto \frac{1}{2\pi} \propto \frac{1}{\sqrt{LC}}$



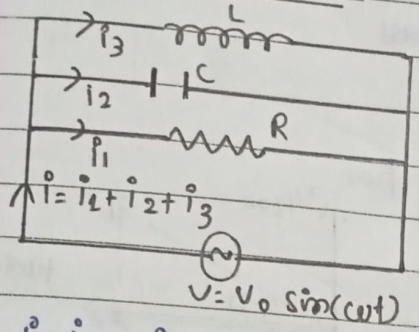

This is desirable, flatter graph means less contrast among sounds of diff. frequencies




negligible



# Parallel RLC circuit



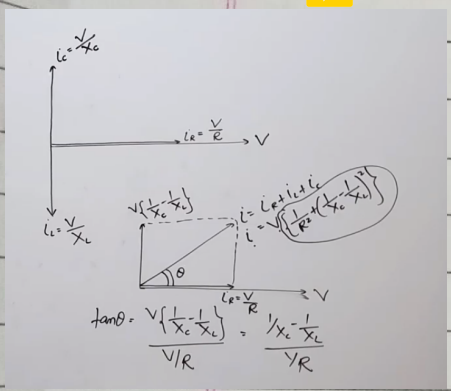

$i_1 = \frac{V_0}{R} \sin(\omega t)$

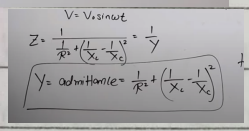
$i_2 = (V_0)\omega C \cos(\omega t) = \left(\frac{V_0}{X_C}\right) \sin(\omega t + \pi/2)$

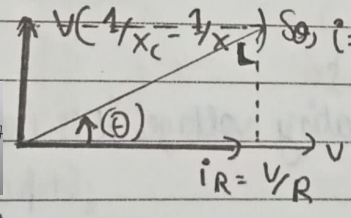
$i_3 = \frac{V_0}{\omega L} \sin(\omega t - \frac{\pi}{2}) = \left(\frac{V_0}{X_L}\right) \sin(\omega t - \pi/2)$

$i = i_1 + i_2 + i_3$

$= \frac{V_0}{R} \sin(\omega t) + \frac{V_0}{X_C} \sin(\omega t + \pi/2) + \frac{V_0}{X_L} \sin(\omega t - \pi/2)$




now,

$1/Z = \sqrt{(\frac{1}{R})^2 + (\frac{1}{X_C} - \frac{1}{X_L})^2}$

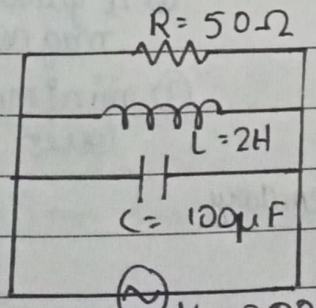
So, $Y$ (admittance) $= \sqrt{(\frac{1}{R})^2 + (\frac{1}{X_C} - \frac{1}{X_L})^2}$

$i(t) = (V_0 Y) \sin(\omega t + \theta)$

$i(t) = (V_0 Y) \sin\left(\omega t + \tan^{-1}\left(\frac{1/X_C - 1/X_L}{1/R}\right)\right)$

$i_{RMS} = \left(\frac{V_0 Y}{\sqrt{2}}\right) = (V_{RMS})(Y)$

Ex-16)




find ① $i_R(t)$ ② $i_L(t)$ ③ $i_C(t)$ ④ Draw phasor ⑤ find admittance and impedance ⑥ $i(t)$ in circuit ⑦ $(i_{RMS})$ in circuit

Soln ① $i_{R(t)} = 4\sin(\omega t) = 4\sin(100t)$ Amperes

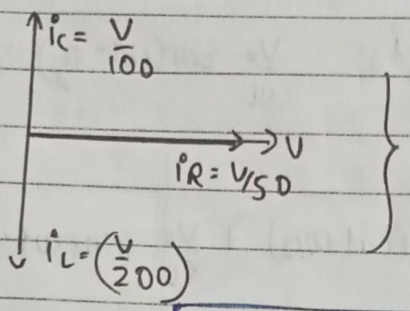
② $i_L(t) = 1\sin(100t - \pi/2)$ Amperes

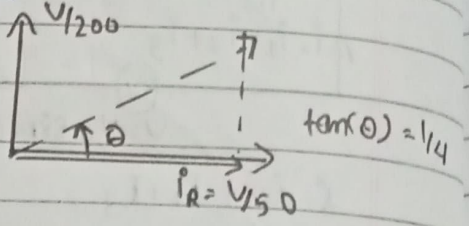
$(X_L = 200\ \Omega)$

③ $(X_C = 100\ \Omega)$

$i_C(t) = 2\sin(100t + \pi/2)$ Ampere

④




⑤ Admittance (Y) $= \sqrt{\left(\frac{1}{200}\right)^2 + \left(\frac{1}{50}\right)^2} = \left(\frac{\sqrt{17}}{200}\right)$ mho

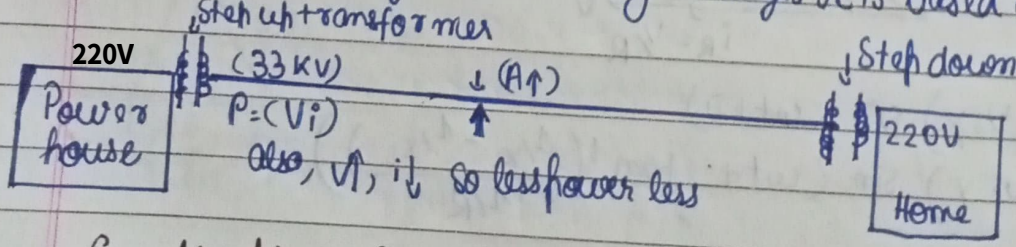
$Z = \left(\frac{200}{\sqrt{17}}\right)$

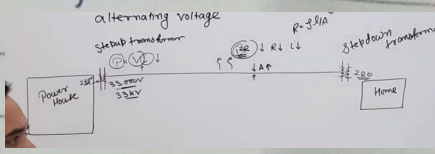
⑥ $I_{(t)} = \sqrt{17}\sin\left(100t + \tan^{-1}\left(\frac{1}{4}\right)\right)$

⑦ $i_{rms} = \sqrt{\frac{17}{2}}$ volts

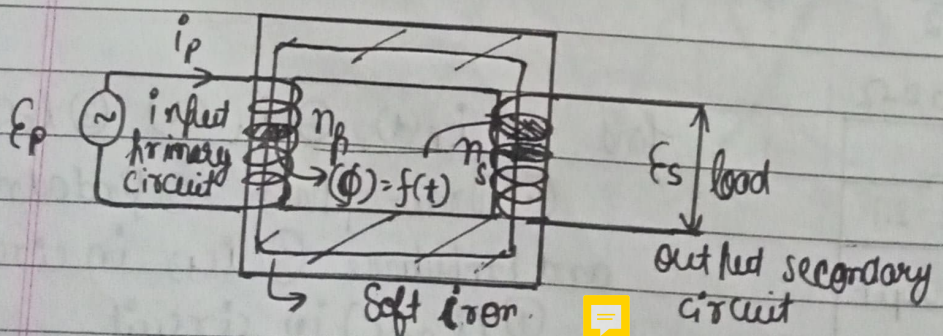
## Transformer (lecture-6)

↳ It is used to change alternating voltage. It is based on mutual induction (induction)



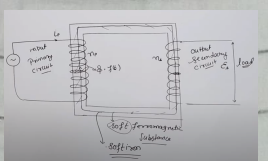

## Concept of transformer




⇒ Core is made of soft iron as:-

① it guides the magnetic field

② minimum hysteresis losses



$\mathcal{E}_p = -\frac{d(n_p \phi)}{dt}$ now, $\mathcal{E}_s = -\frac{d}{dt}(n_s \phi)$

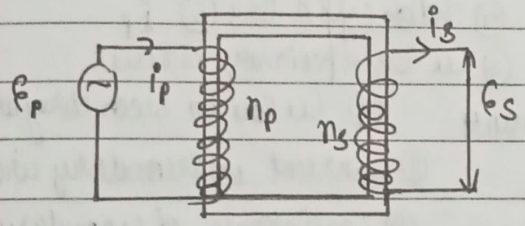
$\mathcal{E}_p = -n_p \frac{d(\phi)}{dt}$ $\mathcal{E}_s = -n_s \frac{d(\phi)}{dt}$

$$\mathcal{E}_s = \frac{n_s}{n_p}(\mathcal{E}_p) \quad \left(\frac{\mathcal{E}_s}{\mathcal{E}_p} = \frac{n_s}{n_p}\right)$$

## Power of ideal transformers

- efficiency = 100%
- $P_{input} = P_{output}$, loss = 0

$\Rightarrow P_s = P_p$



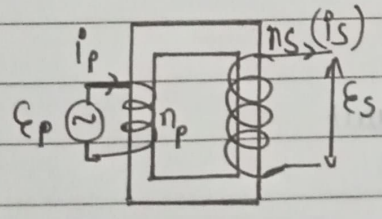

$P_{in} = \mathcal{E}_p i_p$, $P_{out} = \mathcal{E}_s i_s$

$\mathcal{E}_p i_p = \mathcal{E}_s i_s$

$\frac{\mathcal{E}_s}{\mathcal{E}_p} = \frac{i_p}{i_s}$

$$\frac{\mathcal{E}_s}{\mathcal{E}_p} = \frac{n_s}{n_p} = \frac{i_p}{i_s}$$

## Step up transformer

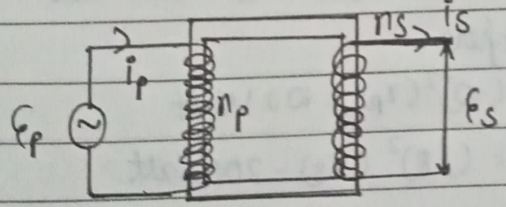



$(n_s > n_p)$

So, $\mathcal{E}_s > \mathcal{E}_p$

Thus $i_s < i_p$

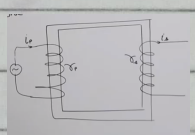
## Step down transformer




$(n_s < n_p)$

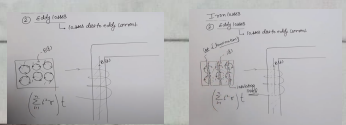
So, $\mathcal{E}_s < \mathcal{E}_p$

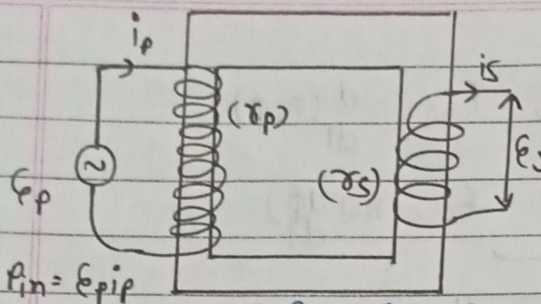
and $i_s > i_p$



## Energy losses in transformers

- copper loss (heating of Cu wire winding, $P_{Cu} = (i_p)^2(r_p) + (i_s)^2(r_s)$)
- iron loss (loss of energy in the soft iron core)

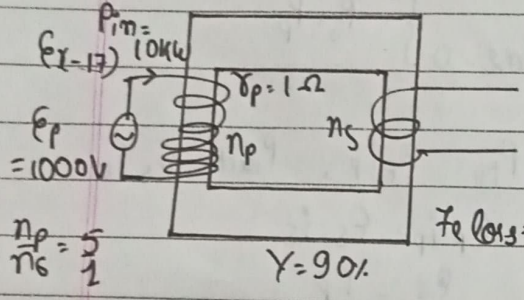





Power loss = $\mathcal{E}_p i_p - \mathcal{E}_s i_s = P_{cu} + P_{fe}$

iron loss = $(\mathcal{E}_p i_p - \mathcal{E}_s i_s) - P_{cu}$

effeciency of transformer = $\left(\frac{P_{out}}{P_{in}}\right) \times 100$




Fe loss = 700 Watt

Find ① $\mathcal{E}_s$ ② $P_{out}$ ③ Total power loss ④ Total copper loss ⑤ $i_p$ ⑥ Cu loss in primary circuit ⑦ Cu loss in secondary circuit ⑧ current in secondary circuit ⑨ resistance of secondary coil

Sol$^n$ ① $\frac{\mathcal{E}_s}{\mathcal{E}_p} = \frac{n_s}{n_p}$

$\mathcal{E}_s = 200$ Volt

② $(90) = \frac{P_{out}}{10KW} \times 100 \Rightarrow P_{out} = 9000$ Watt $= 9$ KW

③ Total power loss = 1000 Watt

④ $P_{cu} + P_{fe} = 1000W \Rightarrow P_{cu} = 300$ Watt

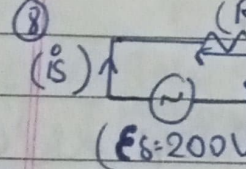
$(i_p)^2 r_p + (i_s)^2 (r_s) = 300$ Watt

⑤ $(i_p) = (10KW)/1KV = 10$ Ampere

⑥ Cu loss in primary circuit = $(10)^2 (r_p) = 100$ Watt

⑦ Cu loss in secondary circuit = $(i_s)^2 (r_s) = 200$ Watt = 300W - 100W

⑧




$P = (\mathcal{E}_s)(i_s) = 9000W + 200W = (9200W)$

$(i_s) = 46$ Ampere is current in secondary circuit

⑨ $(i_s)^2 (R_s) = 200$ Watt

So, $r_s = \left(\frac{200}{(46)^2}\right) \Omega$